

Repsol Electra 3X

Paraffinic dielectric oil — Inhibited

PRODUCT DATA SHEET

Description

ELECTRA oils are light, paraffinic oils designed specifically for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. They are manufactured from highly-refined treated base oils which ensure the lack of any solid, polar or water compounds. The hydrogenation process guarantees that they do not contain any corrosive sulphur, which has been proven to be extremely damaging to transformers.

Classification and standards

ELECTRA 3X is an inhibited oil formulated with antioxidant additives. It complies with the IEC 60296 and ASTM D3487 Type II standards.

Technical specifications

Values according to ASTM D3487 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		ASTM D1524		Bright and clear	
Colour		ASTM D1500		0.5	L0.5
Relative density at 15 °C	g/cm3	ASTM D1298		0.91	0.873
Viscosity at 0 °C	cSt	ASTM D445		76.0	63.96
Viscosity at 40 °C	cSt	ASTM D445		12.0	10.85
Viscosity at 100 °C	cSt	ASTM D445		3.0	2.79
Pour point	°C	ASTM D97		-40	-42
Aniline point	°C	ASTM D611	63		96.4
Interfacial tension	dynes/cm	ASTM D971	40		43
Chemical properties					
Acidity	mg KOH/g	ASTM D974		0.03	<0.01
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Water content	ppm	ASTM D1533		35	19
Inhibitor content	% weight	ASTM D2668		0.3	0.3
Electrical properties					
DDF 60 Hz, 25 °C	%	ASTM D924		0.05	0.009
DDF 60 Hz, 100 °C	%	ASTM D924		0.30	0.194
Breakdown voltage 60 Hz - disk electrodes	kV	ASTM D877	30		61
BDV 60 Hz - VDE 0.04 in (1.02 mm) gap	kV	ASTM D1816	20		35

Properties	Units	Method	Min.	Max.	Typical value
BDV 60 Hz - VDE 0.08 in (2.03 mm) gap	kV	ASTM D1816	35		
BDV impulse, 25 °C, 1-in (25.4 mm) gap	kV	ASTM D3300	145		244
Gassing tendency	uL/min	ASTM D2300		30	15.4
Oxidation stability — Acid-sludge test					
TAN (72 h)	mg KOH/g	ASTM D2440		0.3	<0.01
TAN (164 h)	mg KOH/g	ASTM D2440		0.4	<0.01
Sludge (72 h)	% weight	ASTM D2440		0.1	<0.01
Sludge (164 h)	% weight	ASTM D2440		0.2	<0.01
Rotating bomb test	min	ASTM D2112	195		366
Health, safety and the environment					
Flash point	°C	ASTM D92	145		178
PCB content	ppm	ASTM D4059		Undetectable	

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.